



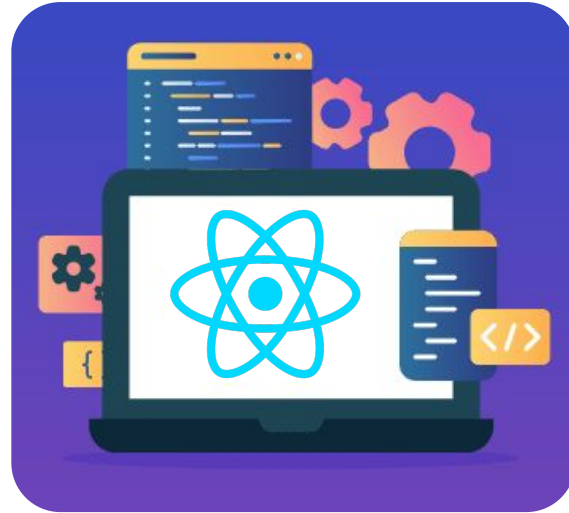
WELCOME





Introduction to React JS

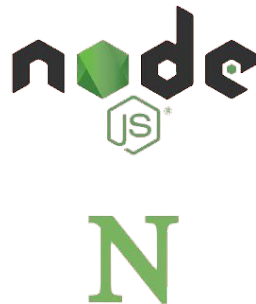
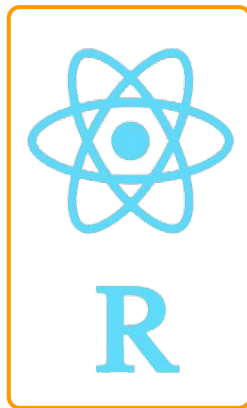
React JS



MERN Stack

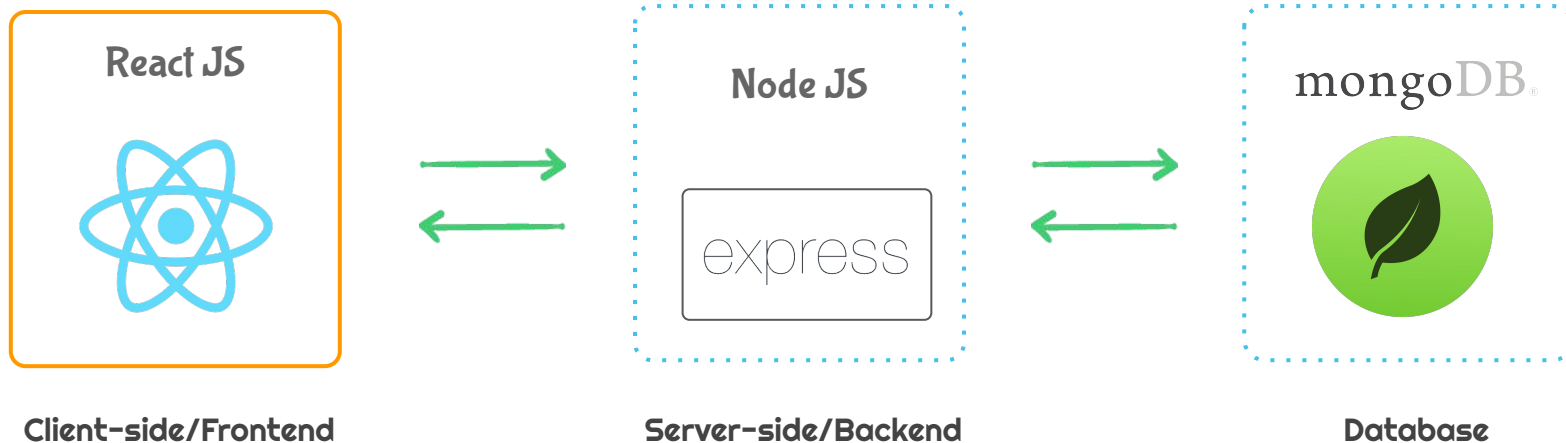
React JS

MERN stands for MongoDB, Express JS, React JS and Node JS



MERN Stack

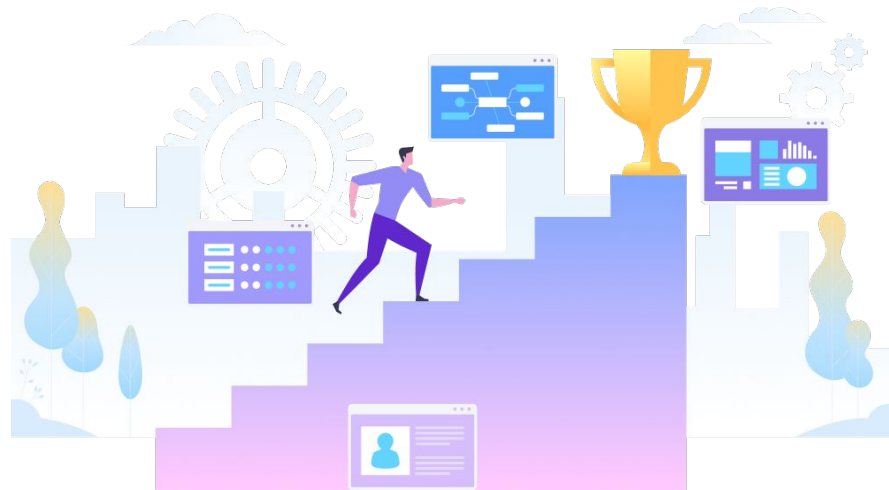
Application Flow



Topics to be Covered

React JS

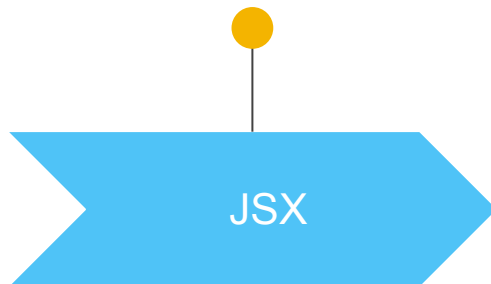
- ✓ Introduction to React & JSX
- ✓ Components and Props
- ✓ Lists and Keys
- ✓ State and Events
- ✓ Handling Side Effects
- ✓ Routing using react-router
- ✓ Authentication & Context



Agenda for Today's Session

Introduction to React JS

Nesting Elements
Embedding Expressions



Introduction

React JS

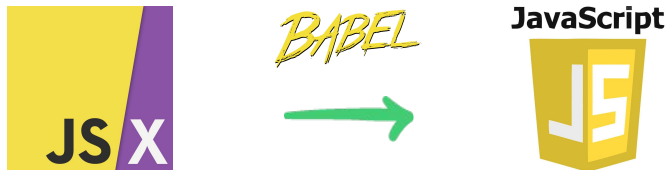
React JS is an **open-source** JavaScript library used to build **user interfaces**



JSX

Babel

A JS compiler that will **translate JSX** into **regular JavaScript**



Create a Heading Element using React JS

Changing Type Attribute

HTML

```
<head>
  ...
  <script src="https://unpkg.com/@babel/standalone@7.27.1/babel.js"></script>
</head>
```

```
<body>
```

```
...
```



```
<script type="module">
```



```
<script type="text/babel" data-type="module">
```

```
  const elementProps = { className: "greeting"};...
```

```
  const element = React.createElement(elementType, elementProps, childrenContent);
```

```
  const root = ReactDOMClient.createRoot(document.getElementById("root"));
```

```
  root.render(element);
```

```
</script>
```

```
</body>
```

Create a Heading Element using React JS

Using JSX

HTML

```
...
<body>
  ...
  <script type="text/babel" data-type="module">...
    - const elementProps = { className: "greeting"};
    - const childrenContent = "Hello World!";
    - const elementType = "h1";
    - const element = React.createElement(elementType, elementProps, childrenContent );
    + const element = <h1 className="greeting">Hello World!</h1>;
    const root = ReactDOMClient.createRoot(document.getElementById("root"));
    root.render(element);
  </script>
</body>
```

Create a Heading Element using React JS

Output

Hello World!

class in HTML and className in JSX

HTML

```
<h1 class="heading"></h1>
```

JSX

```
<h1 className="heading"></h1>
```

Creating a Heading Element

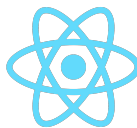
Summary

JavaScript



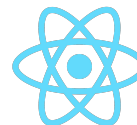
```
document.createElement()
```

```
rootElement.appendChild()
```



```
React.createElement()
```

```
ReactDOMClient.createRoot().render()
```



+



```
<h1>Hello World!</h1>
```

```
ReactDOMClient.createRoot().render()
```

Embedding Variables and Expressions in JSX

Embedding Variables in JSX

Using { }

HTML

```
<body>
  ...
  <div id="root"></div>
  <script type="text/babel" data-type="module">...
    const name = "Rahul";
    const className = "greeting";
    const element = <h1 className="greeting">Hello World!</h1>;
    const element = <h1 className={className}>Hello {name}!</h1>;
    const root = ReactDOMClient.createRoot(document.getElementById("root"));
    root.render(element);
  </script>
</body>
```


Embedding Variables in JSX

Output

Hello Rahul!

Embedding Expressions in JSX

Using { }

HTML

```
<body>
  ...
  <div id="root"></div>
  <script type="text/babel" data-type="module">...
    const fullName = user ⇒ user.firstName + " " + user.lastName;
    const user = { firstName: "Rahul ", lastName: "Attuluri" };
    const element = <h1 className={className}> Hello, {fullName(user)}!</h1>;
    const root = ReactDOMClient.createRoot(document.getElementById("root"));
    root.render(element);
  </script>
</body>
```

Output

Hello, Rahul Attuluri!

Nesting JSX elements

Nesting JSX elements

Example

HTML

```
<body>
  ...
  <script type="text/babel" data-type="module">...
    const element = (
      <div>
        <h1 className="greeting">Hello!</h1>
        <p>Good to see you here.</p>
      </div>
    );
    const root = ReactDOMClient.createRoot(document.getElementById("root"));
    root.render(element);
  </script>
</body>
```

ReactDOMClient.createRoot().render()
returns only **one element** in render

Embedding Expressions in JSX

Output

Hello!

Good to see you here.

Including External JS File in HTML

index.js

index.js

```
import React from 'https://esm.sh/react@19.1.1/?dev'
import ReactDOMClient from 'https://esm.sh/react-dom@19.1.1/client?dev'

const element = (
  <div>
    <h1 className='greeting'>Hello!</h1>
    <p>Good to see you here.</p>
  </div>
)

const root = ReactDOMClient.createRoot(document.getElementById('root'))
root.render(element)
```


Including External JS File in HTML

Syntax

HTML

```
<script type="TYPE_OF_SCRIPT" src="PATH_TO_JS_FILE" ></script>
```

Including External JS File

HTML

```
<body>  
  ...  
  <script type="text/babel" data-type="module" src="index.js"></script>  
</body>
```

Including External JS File

Output

Hello!

Good to see you here.

Including External JS File in HTML

Use only one method

HTML

```
<script type="text/babel" data-type="module" src="index.js"></script>
```

OR

HTML

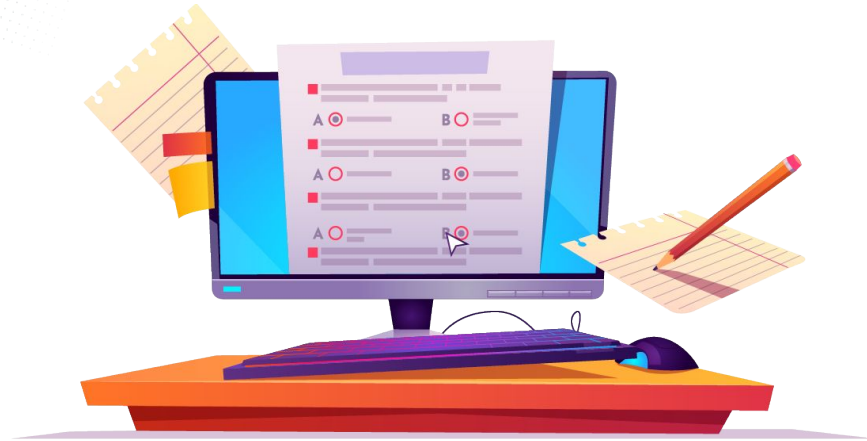
```
<script type="text/babel" data-type="module">
  ...
  const element = <h1 className="greeting">Hello!</h1>
  const root = ReactDOMClient.createRoot(document.getElementById('root'))
  root.render(element)
</script>
```

Key Takeaways

- ✔ JSX
 - Embedding Expressions
 - Nesting elements



Practice



Next Session

Components & Props





THANK YOU





ALL THE BEST

